

A Novel Approach Undergraduate Quantum Mechanics

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I. INTRODUCTION

The standard presentation of quantum mechanics at an undergraduate level relies heavily on rather arbitrary postulates, focuses on wave mechanics, and does not describe the structure of quantum systems, making the subject seem unintuitive and disconnected to students, when in reality, it is one of the most elegant theories ever created. Due to the rather abstract nature of the subject, traditional pedagogy relies on history or Schrodinger's wave mechanics as motivation. Rather than avoiding the abstract nature, the approach described in this manuscript embraces the abstractness, focusing on fundamental theory rather than direct application. This manuscript will serve as a brief conceptual overview of the topics and structure of this novel pedagogical approach. In each section, I will discuss the differences that I have decided on and the reasoning for why I am approaching it differently. The companion paper will act like a shortened textbook for those looking to self-study or apply this approach in the classroom.

II. QUANTUM SYSTEMS AND HILBERT SPACES

In the standard presentation, the subject often delves directly into Schrodinger wave mechanics, leaving the impression that Quantum Mechanics is simply solving integrals. The lack of fundamental structure also makes the subject unintuitive and confusing. For example, the usage of unitary and Hermitian operators is unmotivated and is simply postulated because "it works". Rather, by approaching the subject from the bottom up, providing only the fundamental postulates, we create a deeper understanding of the subject.

A. What is a Quantum System?

When we study quantum systems, it is important to pose the question: what exactly is a quantum system? The answer is rather trivial, but it is important to explicitly state as to motivate the mathematics behind quantum mechanics.

Definition 1. A **Quantum System**, is a physical system in which the effects of quantum mechanics affect the behavior of the system. These effects often only occur on microscopic scales, but emergent effects are common¹.

When we propose a quantum system, we propose its existence separate from any environmental effects. This is analogous to when we propose a classical system in an inertial frame, even though we exist in a gravitational field. We imagine the system existing in a true vacuum separate from any other systems unless otherwise noted. Some examples include a free particle, the hydrogen atom, an electron in a potential, a superconducting magnet, etc.

¹ Such as in condensed matter physics

In classical mechanics, we discuss the behavior of a physical system through its state. For example, for a system with one object, we may discuss the position and momentum of the system at a specific moment in time. This is the state of the physical system. We know from Hamiltonian Mechanics, this state evolves through time governed by Hamilton's Equations based on the constraints and potentials of the system. We call the space in which the state lives, the phase space, through which we can extract all information about the system.

In quantum mechanics, there exists a similar idea of a state, which we can represent through a state vector which we commonly denote as $|\psi\rangle$. This vector can be thought of as containing the position and momentum of the system, but similarly to classical mechanics, we can also extract all information about the system through this state vector including energy, angular momentum, among others. This is the motivation behind our first postulate in Quantum Mechanics

Postulate 1. The state of a quantum system can be fully described through a vector $|\psi\rangle$ in a complex Hilbert Space.

B. Hilbert Spaces

A Hilbert space is a vector space that generalized the concept of Euclidean Space. We will develop the concept of a Hilbert Space by introducing the necessary structure on an abstract vector space and connecting those concepts to standard Euclidean Space. Since the Hilbert space is a complex vector space, in the following definitions, we will take $\mathbb{C}$ to be the field that we are working over. It is easy to take $\mathbb{R}$ to be our field for real vector spaces.

Definition 2. A **Vector Space** V is a set consisting of elements called vectors, with two operations called addition and scalar multiplication such that for any $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and scalar values $a, b \in \mathbb{C}$

1. $\mathbf{u} + \mathbf{v} \in V$
2. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
3. $a\mathbf{v} \in V$
4. INCLUDE ALL VECTOR SPACE AXIOMS

Some examples of vector spaces include: $\mathbb{R}^n$, $\mathbb{C}^n$, the set of all $n \times m$ matrices, the set of all polynomials of degree n , and of course Hilbert spaces $\mathcal{H}$. When we discuss vectors in Euclidean space, we imagine them having a magnitude and direction. This structure is not inherent to the abstract vector space, and thus, we must introduce their notions.

Definition 3. A Norm is a function on a vector space $\|\cdot\| : V \rightarrow \mathbb{R}$, that satisfies the following:

1. $\|\mathbf{v}\| \geq 0$ for any $\mathbf{v} \in V$ and if $\|\mathbf{v}\| = 0$, then $\mathbf{v} = 0$
2. $\|\mathbf{v} + \mathbf{u}\| \leq \|\mathbf{v}\| + \|\mathbf{u}\|$ for any $\mathbf{u}, \mathbf{v} \in V$
3. $\|c\mathbf{v}\| = c\|\mathbf{v}\|$ for any $\mathbf{v} \in V$ and $c \in \mathbb{C}$

The norm that we use in Euclidean space, is the standard Euclidean norm, which for $\mathbb{R}^3$ is defined as $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$. The other generalized notion that we need is that of direction.

Definition 4. An Inner Product is a function on a vector space $\langle \cdot | \cdot \rangle : V \times V \rightarrow \mathbb{C}$ such that the following are satisfied:

1. $\langle u|v \rangle = \overline{\langle v|u \rangle}$
2. $\langle u|av + bw \rangle = a\langle v|u \rangle + b\langle v|w \rangle$
3. $\langle v|v \rangle \geq 0$ and $\langle v|v \rangle = 0$ only if $v = 0$

The inner product in standard Euclidean space is the dot product. Further the dot product induces the standard norm on Euclidean space since $\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$. In fact this is true for any abstract inner product.

Proposition 1. *Given an inner product space, there exists an induced norm by the inner product of the form*

$$\sqrt{\langle \mathbf{v} | \mathbf{v} \rangle}$$

The final structure that we must introduce that is intrinsic to Euclidean space, is the concept of a dual space.

Definition 5. Given a vector space V , a dual space is the set of all linear functions $f : V \rightarrow \mathbb{C}$, denoted as V^*

One can think of this as the set of all the functions that map any vector $\mathbf{v} \in V$ to a scalar value in $\mathbb{C}$. Then this set of functions also form a vector space, isomorphic to the original space V . In fact, the dual space has a special correlation with the inner product in Euclidean space, as well as Hilbert spaces.

Proposition 2. *Given a vector space V and its dual space V^* there exists a correspondence with the linear functionals in V^* and the inner product actions generated by vectors in V . That is, all linear functionals in V^* have the form $f_u(v) = \langle u|v \rangle$ for all $u \in V$*

Then there exists an intimate connection between the norm, the inner product, and the dual space in Euclidean spaces, that we seek to capture in Hilbert Spaces.

Definition 6. A Complex Hilbert Space is a complex inner product space that is also a complete metric space with respect to the norm induced by the inner product.

Note that this definition of the Hilbert Space is not limited to finite dimensions. In fact, we often work with infinite dimensional Hilbert Spaces, including through standard Schrodinger wave mechanics. Then it is beneficial to imagine the Hilbert space as the phase space in which our quantum system lives. That is, the Hilbert space contains all of the possible state vectors for a given quantum system.

C. Notation in Quantum Mechanics

Before we introduce more structure in these Hilbert Spaces, let us define some notation that is often used in presentations of quantum mechanics. Vectors in our Hilbert space are denoted as $|\phi\rangle$ called a ket-vector. Then, dual vectors are denoted with $\langle\phi|$, which we call bra-vectors. This is the bra-ket notation formulated by Dirac, which has the benefit of simplifying notation when we introduce the inner product $\langle\phi|\psi\rangle$. Because of the structure of Hilbert spaces, we can represent the inner product between two vectors as a bra-vector acting on a ket-vector.

Let us discuss a property of our inner product space that will let us connect our bra and ket-vectors.

Proposition 3. Let $\mathcal{H}$ be a Hilbert space, $|\phi\rangle, |\phi'\rangle, |\psi\rangle \in \mathcal{H}$ vectors in the Hilbert space, and $a, b \in \mathbb{C}$ scalars. Then:

$$\langle a\phi + b\phi' | \psi \rangle = \bar{a}\langle \phi | \psi \rangle + \bar{b}\langle \phi' | \psi \rangle$$

This gives us the correspondence $c^*|\phi\rangle \leftrightarrow c|\phi\rangle$. We can imagine the bra-vectors being the complex conjugates of the ket-vectors.

Analogous to linear maps in Euclidean space, there exist linear operators that can act on vectors in our Hilbert space. These operators are linear functions between Hilbert spaces, and in general we work with operators that map vectors into the same Hilbert space. We denote these operators through capital letters A , and their action on ket-vectors as $A|\phi\rangle$. In general, the operator returns a new ket-vector $|\phi'\rangle = A|\phi\rangle$ in the Hilbert space.

In bra-ket notation, operators act on bra-vectors from the right $\langle\phi'| = \langle\phi|A$. Since there is a correspondence between the bra and ket-vectors, we expect that there exists an operator $\langle\phi'| = \langle\phi|B$ such that it is correspondent to $|\phi'\rangle = A|\phi\rangle$. We define the operator $B = A^\dagger$, called the Hermitian adjoint of A . Then with the properties of the inner product, we get the property:

Proposition 4. For vectors $|\phi\rangle, |\psi\rangle \in \mathcal{H}$, and operator $A : \mathcal{H} \rightarrow \mathcal{H}$

$$\langle\phi|A|\psi\rangle = \langle\phi|A^\dagger|\psi\rangle^*$$

Where in bra-ket notation, the operator acts on the ket-vector to the left. That is, $\langle\phi|A|\psi\rangle = \langle\phi|(A|\psi\rangle)$. When we want the operator to act on the bra-vector, we denote it explicitly as: $(\langle\phi|A)|\psi\rangle$.

Let us conclude this section with some further definitions and properties of vectors in Hilbert spaces and operators. These properties are easily derived from the above propositions

Definition 7. A normalized vector ϕ is a vector such that

$$\langle\phi|\phi\rangle = 1$$

Definition 8. Two vectors $|\phi\rangle, |\psi\rangle \in \mathcal{H}$ are called orthogonal if

$$\langle\phi|\psi\rangle = 0$$

Definition 9. Given an operator $A : \mathcal{H} \rightarrow \mathcal{H}$, a vector $|\phi\rangle$ is an eigenvalue of A if:

$$A|\phi\rangle = a|\phi\rangle$$

for some scalar $a \in \mathbb{C}$

Definition 10. We say an operator A is Hermitian if

$$A = A^\dagger$$

Definition 11. We say an operator A is Unitary if

$$AA^\dagger = A^\dagger A = \mathbb{1}$$

Proposition 5. Given a Hilbert space with vectors $\phi \in \mathcal{H}$ and operators $A, B, C : \mathcal{H} \rightarrow \mathcal{H}$:

1. In general, $AB \neq BA$
2. $(AB)C = A(BC)$
3. $A = B$ iff $A|\phi\rangle = B|\phi\rangle$
4. $A + B = B + A$
5. $A + (B + C) = (A + B) + C$
6. $(AB)^\dagger = B^\dagger A^\dagger$

III. SYMMETRIES AND OBSERVABLES

This section provides the background for the fundamental motivation behind the postulate that observables are represented through Hermitian operators. The goal of this chapter is to construct the Hermitian operators through Stone's theorem on one parameter unitary groups. While the complete description requires knowledge in group theory, differential geometry, and functional analysis, we will restrict these topics to a conceptual level so that the text remains accessible to undergraduate students. A deeper dive with proofs will be included in the appendix on the complete version. This, then provides students with a logical line of reasoning from fundamental symmetries into Hermitian observables. While the student still must take theorems for granted, it is more reasonable and intuitive than simply postulating that all observables can be extracted by using an arbitrary Hermitian operator.

A. Introduction

In this section, we will construct the operators associated with observables in an natural manner based solely on fundamental principles. This will require quite a mathematical detour from the previous section, but as we construct the operators, we will develop a deeper and more intuitive understanding of the Hilbert space and how observable phenomenon emerge.

As a brief roadmap, we will discuss symmetries, Lie group, and Lie algebras, before touching on representation theory, and finally ending the construction of our observable operators.

B. Lie Groups and Lie Algebras

In the universe, there exists certain symmetries that the laws of physics remain invariant under. For example, we expect that a ball takes the same time to roll down a ramp at the University of Alabama and at CERN. Similarly, we expect the period a 2D pendulum to be the same no matter what orientation we place it in 3D space

In order to properly discuss these symmetries, we must understand the mathematical language that they are encoded through.

Definition 12. A Group is a set G with an operation $\cdot$ such that for any $x, y, z \in G$:

1. $x \cdot y \in G$
2. $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
3. There exists an element 1 such that $1x = x1 = x$
4. For each element x , there exists an inverse element $x^{-1} \in G$ such that $x \cdot x^{-1} = x^{-1} \cdot x = 1$

It is acceptable to gloss over this definition as we will leave the details of group theory to the mathematicians. It is just important to note that groups are closed under their operation and that inverses exist. The prototypical example for a group is the rotation group, and it is easy to show that rotations satisfy the group axioms.

In fact, the group of rotations belong to a special subset of groups called Lie groups. However, to develop the idea of Lie groups, we first must develop some intuition with manifolds.

Definition 13. An n -dimensional Manifold is a space that locally resembles $\mathbb{R}^n$. That is, for every point on the manifold, there exists a neighborhood around the point that resembles $\mathbb{R}^n$. Further, we say a manifold is Smooth if the manifold is similar enough to $\mathbb{R}^n$ such that one can apply calculus on the manifold.

While this definition is very imprecise, the rigorous definition requires us to delve into topology and differential geometry, which we do not expect in this text. A more rigorous development of these ideas can be found in the appendix of the full text. Some examples of manifolds include: $\mathbb{R}^n$, the surface of a sphere, Minkowski spacetime, curved spacetime, and most importantly Lie groups. However, before we go into the definition of a Lie group, let us first define some relevant concepts related to manifolds.

Definition 14. Given an n -dimensional manifold M , the tangent space at a point p is the set of all tangent vectors at p , denoted as a set $T_p M$.

Note that the tangent space is indeed a vector space, while the manifold is a more general structure. In fact one can show that the tangent space for an n -dimensional manifold is a vector with dimension n . Then now let us discuss the notion of a Lie group.

Definition 15. A Lie group is a smooth manifold G , where G is also a group such that group multiplication and inverses are smooth functions on the manifold.
